

~~### 1. Introduction (Hook & Problem Definition)~~

~~<<Need to quickly narrowing down to the topic of anchoring technique design first>>~~

~~<<then discuss anchoring design's challenges below>>~~

~~Mobile AR applications require users to simultaneously process both physical and digital information while navigating dynamic environments. This creates a fundamental design challenge: ****how to display digital content that allows effective processing of both physical and digital information****. A successful strategy must balance the needs of both domains.~~

~~The challenge is straightforward when users remain stationary—digital content can be positioned to avoid blocking physical content. However, mobility introduces complexity. Users must switch attention between different physical objects (tools, artifacts, instructions) or navigate through entirely changing environments. ****How to place digital content while allowing users to easily access information in the physical environment during attention switching or dynamic movement becomes significantly more complex****.~~

~~This paper investigates the trade-offs between interface accessibility and user field-of-view interference in mobile AR contexts, examining how different anchoring strategies affect user performance across varying cognitive and mobility demands.~~

Augmented Reality overlays digital information onto physical environments to create seamless hybrid experiences that enhance users' understanding and interaction with their surroundings. Central to this integration are anchoring techniques, which manage the spatial relationship between virtual content and the user's reference frames. Three approaches are commonly used in practice: (1) ****world-locked**** anchoring, which fixes content to physical locations, providing stable environmental context but requiring user movement to maintain access; (2) ****head-locked**** anchoring, which attaches content to the user's view direction, ensuring continuous availability but occluding parts of the visual field; and (3) ****body-locked**** anchoring, which positions content relative to the user's torso orientation, balancing accessibility with reduced central occlusion.

While these techniques are simple to understand and work well in many scenarios, they prove less sufficient for the complex demands under mobile AR usage. In mobile contexts, users often need to simultaneously process both physical and digital information while navigating dynamic environments, creating competing demands for visual attention and cognitive resources. The challenge intensifies as task complexity and mobility demands increase: users must dynamically shift attention between different physical objects, navigate through entirely changing surroundings, and maintain access to digital information throughout continuous movement. Traditional anchoring approaches—optimized for either stability (world-locking) or digital information accessibility (head/body-locking)—can become insufficient to adapt to these fluctuating priorities.

Rather than adhering to a single universal policy, optimal anchoring strategy design can depend on contextual factors including user mobility state, environmental complexity, task cognitive

load, and attention switching frequency. This paper investigates the fundamental trade-offs between interface accessibility and field-of-view interference in mobile AR contexts, examining how four anchoring strategies affect user performance across varying cognitive and mobility demands. We developed four techniques that expand beyond basic world/head/body locks: **ScaleStable** (world-locked with distance-based scaling), **PeriphGuide** (head-locked peripheral clamping), **AdaptiveLock** (toggles between world-locked stationary and body-locked walking), and **AdaptiveFlow** (smooth body-to-world transitions).

Through two controlled studies simulating realistic mobile AR usage scenarios, we found important trade-offs in technique effectiveness. Study 1 (individual dual-tasking under basic mobility demands) showed adaptive and body-locked techniques outperformed ScaleStable on interface access and memory recall, suggesting accessibility matters most when cognitive demands are moderate but mobility requirements are high. However, Study 2 (collaborative sensemaking under higher cognitive load but moderate mobility requirements) revealed ScaleStable's stability advantages outweighed accessibility costs, yielding better perceived collaboration despite lower interface attention. Adaptive techniques showed high gaze switching but increased workload during collaborative reasoning.

These results suggest that optimal anchoring requires contextual assessment of the complete visual-cognitive system rather than solutions focused on maximizing any single factor alone.

2. Related Work

[Standard related work section]

3. Design Space & Technical Implementation

2. Design Space & Techniques

3.1 The field of view interference — digital information accessibility Trade-off

Field of view interference: How much digital interface occludes/competes with environmental processing

Digital Information Accessibility: How easily digital content can be accessed during different movement patterns

The mobility factor: Stationary vs. on-the-go changes the equation fundamentally

3.2 Technique Design Principles

ScaleStable: World locked with adaptive scaling/angle → minimizes competition, maintains accessibility within range

PeriphGuide: Head locked peripheral → constant accessibility, low competition in central view

AdaptiveLock: World locked when stationary, body locked when moving → tries to optimize both

AdaptiveFlow: Smooth transitions between states → attempts to minimize cognitive disruption during switching

3.1 The Anchoring Trade-off Space

Mobile AR anchoring involves choosing between two competing strategies: **world-based anchoring** (content fixed in physical space) and **user-based anchoring** (content attached to the user's head (head-lock) or torso (body-lock)). The effectiveness of these strategies seem to relate to a number of factors below.

* **Field-of-view interference**: The degree to which digital interfaces occlude or compete with environmental processing. Body-based anchors keep content consistently positioned relative to the user, often occupying the central visual field and persistently blocking views of physical objects, landmarks, and navigation paths. World-based anchors, in contrast, place content at dedicated environmental locations, minimizing central occlusion but requiring users to consciously look away from their current task to access information.

* **Digital information accessibility**: How readily users can access digital content during different movement patterns. Body-based anchors provide constant availability—content remains at a fixed relative position regardless of user movement. World-based anchors, however, become progressively less accessible as users move: content fixed in the world becomes harder to locate, may fall outside the field of view, and can become illegible due to distance scaling if not dynamically adapted.

The **mobility factor** also heavily influences this trade-off equation. When stationary, users can afford to allocate attention to peripheral or distant world-anchored content with minimal performance cost, benefiting from reduced central occlusion. However, during movement—whether walking, turning, or navigating—the cognitive bandwidth for seeking out distant content decreases while environmental processing demands increase. World-anchored content becomes not just less accessible but actively disruptive as users must interrupt navigation to locate and orient toward it. Body-anchored content remains accessible during movement but creates persistent occlusion exactly when environmental awareness is most critical.

This creates distinct operational regimes: stationary periods favor world-anchoring's reduced interference, while mobility periods favor body-anchoring's accessibility. Traditional anchoring approaches often need to trade-off one regime over the other, leading us to explore alternative techniques that overcome some of their limitations based on user state.

3.2 Technique Design Principles

Guided by this trade-off framework, we developed four alternative anchoring techniques that occupy distinct positions in the design space:

ScaleStable (world-locked with adaptive scaling/angle) addresses accessibility limitations of traditional world-locking by dynamically scaling content based on user distance. As the user moves farther away, the interface expands to maintain legibility; as they approach, it shrinks to avoid excessive occlusion. Continuous user-facing rotation ensures readability from any

approach direction. This approach ****minimizes central field competition while maintaining accessibility within a defined operational range**** (0.5–3.5 m).

****PeriphGuide**** (head-locked peripheral clamping) optimizes for constant accessibility by keeping content within the user's view at all times. However, unlike traditional head-locking, content is constrained to the peripheral visual field (clamped to $\pm 40^\circ$ horizontal, $\pm 30^\circ$ vertical) through view-clamping logic. This ensures ****continuous access while limiting competition with central visual tasks****, effectively creating a "radar-like" interface that remains available without demanding foveal attention.

****AdaptiveLock**** (world-locked when stationary, body-locked when moving) attempts to dynamically optimize for both regimes. When the user is stationary (>0.5 s without movement), content anchors to a world position, providing spatial stability. Once walking begins (>0.3 m/s), the system transitions to body-locking, keeping content accessible during navigation. This approach ****tries to provide optimal positioning for each mobility state**** but introduces explicit state transitions that users must cognitively track.

****AdaptiveFlow**** (smooth transitions between states) addresses the cognitive disruption costs of abrupt switching. Rather than binary toggling between world and body locks, this technique uses gradual interpolation over ~ 2 -second windows. During movement, content smoothly migrates from body-locked positions to peripheral world positions, then glides back as motion subsides. This ****attempts to minimize cognitive disruption during state changes**** by maintaining continuous spatial trajectories that users can more easily anticipate and track.

All four techniques were implemented on HoloLens 2 with Unity, using built-in spatial anchors and spatial awareness systems. We employed temporal smoothing (Move/Rotate Lerp = 0.1) to mitigate tracking jitter, and implemented empirical motion thresholds (stationary: <0.3 m/s; slow: 0.3 – 1.5 m/s; fast: >1.5 m/s) to drive adaptive transitions in AdaptiveLock and AdaptiveFlow.

~~### 4. Study 1: Interface Performance Under Basic Mobility Demands~~

~~#### 4.1 Motivation and Hypotheses~~

~~Study 1 examines interface performance under controlled mobility with information processing. We aimed to understand how anchoring techniques affect individual performance in scenarios requiring both spatial navigation and information retention—a foundational dual-tasking context where users must coordinate environmental awareness with interface access. We anticipated that under these basic conditions, users might prioritize accessibility over perfect spatial stability, making adaptive techniques potentially advantageous. However, given the relatively simple cognitive task (memory recall), we also expected traditional world-locked anchoring to perform adequately when users were stationary.~~

~~**H1**~~: Under basic dual-tasking demands, body-locked and adaptive techniques would provide greater interface accessibility during movement, leading to more interface viewing time compared to purely world-locked approaches.

~~**H2**~~: Techniques with continuous access (body-locked and adaptive) would increase gaze switching frequency, reflecting more dynamic attention allocation during walking phases.

~~**H3**~~: Despite greater accessibility, adaptive techniques with visible transitions might incur subjective costs in temporal availability or spatial consistency.

<<Need to change the above to research questions instead of hypothesis>>

~~### 4.2 Method~~

~~#### Participants~~

We recruited 12 participants (P1-P12; 10 male, 2 female; age 21–30, M=24.3) with normal or corrected-to-normal vision and basic smartphone experience.

~~#### Task Design~~

Participants performed two concurrent tasks simulating navigating while retrieving collaborative history:

~~**Mobility Task: Waypoint Click.**~~ To simulate dynamic walking conditions, we designed a navigation task featuring four fixed waypoints labeled 1–4. In each 5-minute condition, numbers 1–4 appeared six times in random order at the bottom of the slide over a 50-second interval. Participants walked to corresponding waypoints via any route and confirmed arrival by clicking a world-locked blue button.

~~**Cognitive Processing Task: Slides Memory.**~~ To simulate recalling short-term collaborative information, participants viewed a six-slide deck for 5 minutes (50 s per slide). Each slide showed two items—numbers, English words, or simple shapes—in a fixed sequence. After each condition, they completed a 12-item immediate recall test (4 per category) to isolate how anchoring affected encoding during movement.

~~#### Interface Conditions~~

Participants experienced all four techniques (ScaleStable, PeriphGuide, AdaptiveLock, AdaptiveFlow) across three physical environments (Narrow Corridor, Room-to-Room, Open Space). Technique is counterbalanced using latin square, the environment is presented in fixed order as we are not investigating how different environments compare with each other.

~~#### Measures~~

~~—**Interface Attention Ratio**~~: Proportion of visual attention to meeting interface (Document panel)

~~—**Gaze Transition Rate**~~: Frequency of gaze shifts between interface AOs and environment, computed separately for walking and stationary segments

~~—**Mean Fixation Duration**~~: Average fixation time on interface elements

~~—**Immediate Recall**~~: Accuracy on the 12-item memory test

~~—**Subjective Ratings**~~: Temporal availability, spatial consistency, workload (NASA TLX), and usability (SUS)

4.3 Results

RQ1: Visual Attention Patterns

ScaleStable yielded significantly less interface viewing time (38.63%) than AdaptiveFlow (54.52%), AdaptiveLock (55.85%), or PeriphGuide (57.79%; all $p < .008$), with a large technique main effect ($F_{3,33} = 12.38$, $p < .001$, $\eta^2 = 0.530$). This supported H1—world-locked anchoring indeed limited access during movement. <<don't use a hypothesis here. Change to research question>>

Gaze transition rates revealed adaptive techniques increased switching over static ones. During walking, AdaptiveLock showed the highest rate (59.63/min), significantly exceeding ScaleStable (31.28/min), supporting H2 <<again, don't use hypothesis, but use research questions>>. AdaptiveLock also produced the most transitions in Open Space, while ScaleStable was lowest in the Narrow Corridor, showing environment-dependent effects.

Mean fixation duration was dominated by technique ($F_{3,33} = 606.40$, $p < .001$, $\eta^2 = .982$): ScaleStable (6509.34 ms) >> AdaptiveFlow (1781.47 ms), PeriphGuide (1943.36 ms), AdaptiveLock (1599.74 ms; all $p < .001$). This indicates that when users could access the world-locked interface, they engaged in longer, more sustained reading.

RQ2: Memory Under Dual-Tasking

ScaleStable significantly impaired recall (8.69 items) versus AdaptiveFlow (10.94), AdaptiveLock (10.56), and PeriphGuide (10.25; all $p < .030$), demonstrating that reduced interface access directly affected cognitive task performance. No environment effects emerged, suggesting technique differences persisted across spaces.

RQ3: Subjective Experience

As expected, PeriphGuide scored highest on temporal availability (5.71), followed by AdaptiveLock (5.56), AdaptiveFlow (5.50), and ScaleStable (5.08), partially supporting H3. Spatial consistency showed only environment effects: Narrow Corridor (5.04) < Open Space (5.98), Room to Room (5.99), suggesting physical constraints mattered more than technique for perceived stability. <<do we worry about spatial consistency any more, I think we don't use that framing>>

Workload (NASA-TLX) was driven by environment: Room-to-Room (62.88) > Narrow Corridor (48.22) > Open Space (36.74), with no technique effects. Usability ratings showed AdaptiveLock (83.75) and ScaleStable (80.83) scored highest on SUS, while preference rankings showed AdaptiveLock/ScaleStable took 92% first-place votes, revealing a subjective preference for techniques offering either perfect stability or clear adaptive signaling.

4.4 Study 1 Discussion

Study 1 confirmed that under basic dual-tasking, continuous interface accessibility provided by body-locked and adaptive techniques improved both objective performance (interface attention, memory recall) and subjective experience (temporal availability). ScaleStable's world-locked positioning created an accessibility gap during movement, even though it allowed longer reading sessions when accessible.

However, interesting tensions emerged: while AdaptiveLock showed the highest gaze-switching and good memory performance, participants still preferred it and ScaleStable over other techniques. This suggests users valued either consistent availability (AdaptiveLock) or perfect stability (ScaleStable), with intermediate approaches like AdaptiveFlow perceived as less usable.

The absence of technique effects on workload indicates that under moderate multitasking demands, accessibility benefits did not come with significant cognitive cost penalties. Yet, these simple memory tasks couldn't reveal how techniques would fare when environmental processing itself became complex—a gap addressed in Study 2. <<raise this as a question, what if the cognitive demand further increase?>>

5. Study 2: Interface Performance Under Collaborative Cognitive Load

5.1 Motivation and Hypotheses <<again, don't use hypothesis, but use research questions>>

Study 2 examines how anchoring techniques affect performance when environmental processing demands increase dramatically—specifically in collaborative sensemaking that requires constant attention switching between physical space, interface content, and partner communication. We designed a dyadic murder mystery investigation requiring navigation between physical evidence stations while synthesizing information with a remote partner. This context tests whether accessibility advantages observed in Study 1 persist when environmental complexity demands sustained visual attention and collaborative reasoning.

****H4**:** Under high collaborative load, world-locked interfaces (ScaleStable) would support better shared spatial reference, improving collaborative outcomes despite reduced accessibility.

****H5**:** Adaptive techniques might increase gaze-switching and communication focus but risk disrupting shared attention and increasing perceived workload.

~~**H6**~~: The trade-off between accessibility and stability would manifest most clearly in subjective collaboration quality and attention coordination metrics.

~~### 5.2 Method~~

~~#### Participants~~

~~12 dyads (24 participants total, 13 female, 11 male, ages 22-38, M=27.8, SD=4.6) with no prior experience in Study 1.~~

~~#### Task Design~~

~~Dyads role-played a murder-mystery investigation requiring collaborative sensemaking. The wearer navigated between two physical stations (3.2 m apart) checking evidence consistency, while the remote partner reviewed digital profiles. The task demanded constant attention shifts among:~~

- ~~1. **Environmental processing**: Examining physical evidence items at stations~~
- ~~2. **Interface interaction**: Accessing digital information and video communication~~
- ~~3. **Partner coordination**: Discussing findings and building shared understanding~~

~~Dyads created a shared Reasoning Brief including event timeline, suspect identification, evidence items, and conclusion. Five equivalent scripts prevented learning effects.~~

~~#### Procedure~~

~~After role assignment and practice, each dyad completed four counterbalanced sessions (one per technique) with questionnaires and interviews. Sessions averaged 23 minutes; full experiments lasted 3–3.5 hours per dyad with remote monitoring.~~

~~#### Measures~~

- ~~**Dyadic attention coordination**: Interface attention ratio, gaze transition rate (walking/stationary), communicative focus ratio (Video vs. Document attention)~~
- ~~**Subjective experience**: Spatial consistency, temporal availability, perceived attention allocation, workload (NASA-TLX), Preference Ranking~~
- ~~**Collaboration outcomes**: Perceived collaboration quality~~
- ~~**Qualitative feedback**: Post-experiment interviews on technique trade-offs~~

~~### 5.3 Results~~

~~#### RQ2a: Dyadic Attention Coordination~~

~~ScaleStable significantly reduced interface attention ratio (25.32%) compared to AdaptiveFlow (40.63%, $p = .003$) or PeriphGuide (40.22%, $p = .003$), with AdaptiveLock intermediate (31.80%). This confirmed that world-locking limited access even more dramatically in complex environments.~~

During walking, AdaptiveFlow produced the highest gaze transition rate (164.26/min), exceeding all others ($p \leq .001$). Stationary rates followed similar patterns: AdaptiveFlow (130.14/min) > others. This indicates that smooth adaptive transitions prompted the most dynamic gaze behavior.

ScaleStable yielded the lowest communicative focus ratio (0.20) versus AdaptiveFlow (0.36), PeriphGuide (0.40), and AdaptiveLock (0.38; all $p < .013$), showing world locking reduced attention to communicative (video) content relative to task (document) content.

RQ2b: Subjective Experience

Despite its accessibility limitations, ScaleStable scored highest on spatial consistency/temporal availability (5.91) > AdaptiveFlow (4.32, $p = .011$), supporting H4. Perceived attention allocation also favored static techniques: AdaptiveFlow (3.84) < AdaptiveLock (5.13, $p = .012$), with ScaleStable descriptively highest (5.46).

Workload differed significantly: AdaptiveLock (38.60) < AdaptiveFlow (58.62, $p = .045$), showing that discrete adaptive switching created lower perceived demand than continuous adaptation.

RQ2c: Collaboration Outcomes

ScaleStable boosted perceived collaboration (6.52) > AdaptiveFlow (6.30, $p = .044$) and PeriphGuide (6.32, $p = .004$), despite—or perhaps because of—its reduced accessibility. This counterintuitive finding suggests that shared spatial stability might outweigh individual access for collaborative quality.

Qualitative Insights

Interviews revealed technique trade-offs clearly. ScaleStable's stability aided predictability: "The fixed location felt stable, and I always knew where it was (W11)"; "Easier to discuss, stayed aligned on the document (W6)." AdaptiveFlow distracted: "Panel always on the move... chasing it with my eyes (W5)"; "Harder to stay in sync (W12)." AdaptiveLock reduced fatigue: "Less tiring... easier to manage (W10)."

5.4 Study 2 Discussion

Study 2 revealed a dramatic context shift: under high collaborative load, ScaleStable's world-locked positioning—despite reducing interface access—enhanced perceived collaboration quality and spatial consistency. The accessibility advantages of adaptive techniques observed in Study 1 were counterbalanced by increased cognitive fragmentation and reduced shared spatial reference.

Three key insights emerge:

1. ****Collaboration prioritizes shared reference over individual access****: ScaleStable's fixed location created a reliable spatial anchor that supported mutual understanding, even at the cost of requiring physical repositioning for access.

2. ~~**Adaptive behavior has cognitive costs**~~: AdaptiveFlow's continuous transitions increased gaze switching and subjective workload without improving collaboration outcomes, suggesting that fluid adaptation might overwhelm users during complex tasks.

3. ~~**Discrete versus continuous adaptation matters**~~: AdaptiveLock's clear state transitions (world ↔ body) were perceived as less demanding than AdaptiveFlow's smooth interpolation, indicating that predictable mode changes might be preferable to continuous adaptation.

The reversal from Study 1—where accessibility drove performance—to Study 2—where stability enhanced collaboration—demonstrates that optimal anchoring depends fundamentally on task context. Simple dual tasking favors continuous access; complex collaboration favors shared spatial consistency.

~~### 5.5 Cross-Study Synthesis~~

Comparing both studies reveals a fundamental tension in mobile AR anchoring:

~~**Low-Complexity Contexts (Study 1)**~~: When environmental processing is simple (navigation + memory), continuous interface accessibility dominates. Adaptive and body-locked techniques improve performance by reducing search costs during movement.

~~**High-Complexity Contexts (Study 2)**~~: When environmental processing itself becomes complex (evidence examination + collaborative reasoning), shared spatial stability and predictable reference frames become paramount. World-locked anchoring supports better collaboration despite accessibility limitations.

This synthesis suggests that no single technique optimizes all contexts. Instead, effective AR interfaces might need to adapt not just to user motion but to task complexity—prioritizing accessibility during simple activities and stability during complex collaboration.

4. Study 1: Interface Performance Under Basic Mobility Demands

4.1 Motivation and Research Questions

Study 1 examines interface performance under controlled mobility with information processing. We aimed to understand how anchoring techniques affect individual performance in scenarios requiring both spatial navigation and information retention—a common dual-tasking context where users must coordinate environmental awareness with interface access.

Given the relatively simple environmental processing demands (waypoint navigation) and moderate cognitive load (memory recall), we expected that digital information accessibility would be the primary constraint on performance. However, we also anticipated potential trade-offs between accessibility benefits and cognitive costs from adaptive transitions.

<<replace with 1 or 2 higher level questions derived from discussion>>

4.2 Method

Participants

We recruited 12 participants (P1-P12; 10 male, 2 female; age 21–30, $M=24.3$) with normal or corrected-to-normal vision and basic smartphone experience.

Task Design

Participants performed two concurrent tasks simulating navigating while retrieving collaborative history:

Mobility Task: Waypoint Click. To simulate dynamic walking conditions, we designed a navigation task featuring four fixed waypoints labeled 1–4. In each 5-minute condition, numbers 1–4 appeared six times in random order at the bottom of the slide over a 50-second interval. Participants walked to corresponding waypoints via any route and confirmed arrival by clicking a world-locked blue button at the designated location.

Cognitive Processing Task: Slides Memory. To simulate recalling short-term collaborative information, participants viewed a six-slide deck for 5 minutes (50 s per slide). Each slide showed two items—numbers, English words, or simple shapes—in a fixed sequence. After each condition, they completed a 12-item immediate recall test (4 per category) to isolate how anchoring affected encoding during movement.

Interface Conditions

Participants experienced all four techniques (ScaleStable, PeriphGuide, AdaptiveLock, AdaptiveFlow) across three physical environments (Narrow Corridor, Room-to-Room, Open Space). Technique order was counterbalanced using latin square; environments were presented in fixed order as we focused on technique comparison rather than environment effects.

Measures

- **Interface Attention Ratio:** Proportion of visual attention to meeting interface (Document panel)
- **Immediate Recall:** Accuracy on the 12-item memory test
- **Subjective Ratings:** Temporal availability, workload (NASA TLX), and usability (SUS)

4.3 Results

<<suggest to re-organize results based on the key discussion points we have later. I think we can focus on the key findings and insights and organize the results to support the main arguments, mobility does affect scalestable's viewing time, and it thus also affects the memory recall, however, having a fixed place to view information seems to be subjectively preferred as shown the SUS ratio.→>>

Cognitive Performance Under Mobility

ScaleStable significantly impaired recall (8.69 items) versus AdaptiveFlow (10.94), AdaptiveLock (10.56), and PeriphGuide (10.25; all $p < .030$). This demonstrates that reduced interface accessibility during mobility directly affected cognitive task performance, despite the higher quality engagement when content was accessible.

Visual Attention Patterns During Mobility

ScaleStable yielded significantly less interface viewing time (38.63%) than AdaptiveFlow (54.52%), AdaptiveLock (55.85%), or PeriphGuide (57.79%; all $p < .008$), with a large technique main effect ($F_{3,33} = 12.38$, $p < .001$, $\eta^2 = 0.530$). This confirmed that world-locked anchoring created substantial accessibility limitations during movement phases.

Subjective Experience and Trade-offs

As expected, PeriphGuide scored highest on temporal availability (5.71), followed by AdaptiveLock (5.56), AdaptiveFlow (5.50), and ScaleStable (5.08), reflecting users' awareness of accessibility differences. Workload (NASA-TLX) was primarily driven by environment rather than technique: Room-to-Room (62.88) > Narrow Corridor (48.22) > Open Space (36.74).

Interestingly, usability ratings showed AdaptiveLock (83.75) and ScaleStable (80.83) scored highest on SUS, while preference rankings revealed AdaptiveLock/ScaleStable received 92% of first-place votes. This suggests users valued either consistent availability (AdaptiveLock) or perfect stability when accessible (ScaleStable), with intermediate approaches like AdaptiveFlow perceived as less optimal.

4.4 Study 1 Discussion

Study 1 confirmed that under basic mobility and moderate cognitive demands, **digital information accessibility seems to be the main reason that affects memory performance.** Techniques providing continuous access (body-locked and adaptive) improved both objective performance (memory recall) and enabled more dynamic attention allocation, supporting effective dual-tasking.

Interestingly, despite the performance penalty, scalestable has a higher SUS score as compared with periphguide and adaptiveflow, which shows users like the fixed anchor point ...

The absence of technique effects on workload indicates that under moderate multitasking demands, accessibility techniques did not impose significant cognitive costs. What about more complex tasks?—raising the question of whether accessibility advantages would persist under higher cognitive loads.

5. Study 2: Interface Performance Under Collaborative Cognitive Load

5.1 Motivation and Research Questions

Study 2 examines how anchoring techniques perform when both environmental processing demands and cognitive load increase dramatically. We designed a collaborative sensemaking task requiring constant attention switching between complex physical evidence examination, digital information synthesis, and partner communication.

This context tests whether the accessibility advantages observed in Study 1 persist when users face **intensive environmental processing demands** (detailed evidence inspection), **high cognitive load** (collaborative reasoning), and **frequent attention switching** between multiple information sources. We anticipated that the increased complexity might shift the optimal balance between accessibility and stability.

RQ2a: How do anchoring techniques affect attention coordination in dyadic collaboration requiring complex environmental processing?

RQ2b: Do cognitive disruption costs from adaptive techniques become more significant under high collaborative load?

RQ2c: How do accessibility-stability trade-offs affect perceived collaboration quality and shared spatial reference?

5.2 Method

Participants

12 dyads (24 participants total, 13 female, 11 male, ages 22–38, $M=27.8$, $SD=4.6$) with no prior experience in Study 1.

Task Design

Dyads role played a murder mystery investigation requiring collaborative sensemaking. The wearer navigated between two physical stations (3.2 m apart) checking evidence consistency,

while the remote partner reviewed digital profiles. The task demanded constant attention shifts among:

1. **Complex environmental processing:** Detailed examination of physical evidence items requiring sustained visual attention
2. **Digital information synthesis:** Accessing and processing multiple information sources through the interface
3. **Collaborative reasoning:** Discussing findings, building shared understanding, and coordinating investigation strategy

Dyads created a shared Reasoning Brief including event timeline, suspect identification, evidence items, and conclusion. Five equivalent scripts prevented learning effects.

Procedure

After role assignment and practice, each dyad completed four counterbalanced sessions (one per technique) with questionnaires and interviews. Sessions averaged 23 minutes; full experiments lasted 3–3.5 hours per dyad with remote monitoring.

Measures

- **Dyadic attention coordination:** Interface attention ratio, communicative focus ratio (Video vs. Document attention)
- **Subjective experience:** perceived attention allocation, workload (NASA-TLX)
- **Collaboration outcomes:** Perceived collaboration quality
- **Qualitative feedback:** Post-experiment interviews on technique trade-offs

5.3 Results

Collaboration Quality and Shared Reference

ScaleStable achieved the highest perceived collaboration quality (6.52) compared to AdaptiveFlow (6.30, $p = .044$) and PeriphGuide (6.32, $p = .004$). This counterintuitive finding suggests that **shared spatial stability outweighed individual accessibility benefits for collaborative performance**.

ScaleStable also scored highest on spatial consistency/temporal availability (5.91) > AdaptiveFlow (4.32, $p = .011$), indicating that users valued predictable interface positioning for collaborative coordination, even when it required physical repositioning for access.

RQ2a: Attention Coordination Under Complex Environmental Demands

ScaleStable significantly reduced interface attention ratio (25.32%) compared to AdaptiveFlow (40.63%, $p = .003$) or PeriphGuide (40.22%, $p = .003$), with AdaptiveLock intermediate

(31.80%). This confirmed that accessibility limitations became even more pronounced when environmental processing required sustained visual attention.

RQ2b: Cognitive Disruption Costs Under High Load

Unlike Study 1, significant workload differences emerged: AdaptiveLock (38.60) < AdaptiveFlow (58.62, $p = .045$), indicating that **adaptive techniques imposed cognitive costs that became apparent under high task complexity**. Discrete mode switching (AdaptiveLock) was less demanding than continuous adaptation (AdaptiveFlow).

Perceived attention allocation also favored static techniques: AdaptiveFlow (3.84) < AdaptiveLock (5.13, $p = .012$), with ScaleStable descriptively highest (5.46), suggesting users felt less control over their attention with continuous adaptive behavior.

Qualitative Insights: Stability vs. Accessibility Trade-offs

Interviews revealed clear technique preferences based on collaboration needs. ScaleStable's stability aided predictability and shared reference: "The fixed location felt stable, and I always knew where it was (W11)"; "Easier to discuss, stayed aligned on the document (W6)."

AdaptiveFlow created distraction during complex reasoning: "Panel always on the move... chasing it with my eyes (W5)"; "Harder to stay in sync when it kept moving (W12)."

AdaptiveLock provided a middle ground: "Less tiring than the moving one... easier to manage when I needed to focus on evidence (W10)."

5.4 Study 2 Discussion

Study 2 revealed a context-dependent shift in optimal anchoring strategy. Under a different setting, high collaborative load and complex environmental processing, ScaleStable achieved the highest performance.

The findings demonstrate three critical insights about **visual cognitive balance under higher complexity**:

1. **Environmental processing complexity changes accessibility priorities:** When physical evidence examination required sustained visual attention, the cognitive costs of visual field competition from continuously accessible interfaces outweighed their benefits.
2. **Cognitive disruption costs emerge under high load:** Adaptive techniques that showed no workload penalties in Study 1 created significant cognitive burden when environmental and collaborative demands increased. Continuous adaptation (AdaptiveFlow) was more disruptive than discrete mode switching (AdaptiveLock).

5.5 Cross-Study Synthesis: Context-Dependent Optimization

- 1) ScaleStable and AdaptiveLock appear to perform well in the two scenarios where AdaptiveLock achieves a higher performance in a simpler cognitive multitasking, but higher mobility-demanding scenario, while scalestable performs better in a more cognitively complex, yet slightly less mobility-demanding scenario.
- 2) PeriphGuide provides always available interface accessibility, but its interference to the user's field of view is always present, thus seems to be less favored by participants as compared with the other two techniques.
- 3) AdaptiveFlow is similar to AdaptiveLock except it has a 2-second transition between modes. Our two studies showed that such transitions produce more harm than help as it creates greater interference for users.

Moderate Cognitive and environmental Demands + higher mobility demands (Study 1): When mobility demands are higher, but navigation is relatively straightforward and cognitive tasks are moderate, **accessibility optimization's benefit outweighs the user field of view interference and potential cognitive interruption.**

High Cognitive and environment Demands + moderate mobility demands (Study 2): When environmental processing requires sustained attention and cognitive tasks involve collaboration, **stability and shared reference's advantage outweighs the loss in accessibility**

This synthesis suggests that effective mobile AR systems may need to move beyond simple motion-based adaptation toward **context-aware optimization** that considers environmental complexity, cognitive demands, and mobility requirements.

<Also I think we need to emphasize the importance of user field of view interference, this is something need to be more serious considered in design, lessons from adaptiveflow, and periphguide>

4. Study 1: Interface Performance Under Basic Mobility Demands

4.1 Motivation and Research Questions

Study 1 examines interface performance under controlled mobility with information processing. We aimed to understand how anchoring techniques affect individual performance in scenarios requiring both spatial navigation and information retention—a common dual-tasking context where users must coordinate environmental awareness with interface access.

Given the relatively simple environmental processing demands (waypoint navigation) and moderate cognitive load (memory recall), we expected that digital information accessibility would be the primary constraint on performance. However, we also anticipated potential trade-offs between accessibility benefits and cognitive costs from adaptive transitions.

RQ1: How does the accessibility of AR information display techniques influence user performance and experience in multitasking scenarios with moderate cognitive and mobility demands?

4.2 Method

Participants

We recruited 12 participants (P1-P12; 10 male, 2 female; age 21–30, $M=24.3$) with normal or corrected-to-normal vision and basic smartphone experience.

Task Design

Participants performed two concurrent tasks simulating navigating while retrieving collaborative history:

Mobility Task: Waypoint Click. To simulate dynamic walking conditions, we designed a navigation task featuring four fixed waypoints labeled 1–4. In each 5-minute condition, numbers 1–4 appeared six times in random order at the bottom of the slide over a 50-second interval. Participants walked to corresponding waypoints via any route and confirmed arrival by clicking a world-locked blue button at the designated location.

Cognitive Processing Task: Slides Memory. To simulate recalling short-term collaborative information, participants viewed a six-slide deck for 5 minutes (50 s per slide). Each slide showed two items—numbers, English words, or simple shapes—in a fixed sequence. After each condition, they completed a 12-item immediate-recall test (4 per category) to isolate how anchoring affected encoding during movement.

Interface Conditions

Participants experienced all four techniques (ScaleStable, PeriphGuide, AdaptiveLock, AdaptiveFlow) across three physical environments (Narrow Corridor, Room-to-Room, Open Space). Technique order was counterbalanced using latin-square; environments were presented in fixed order as we focused on technique comparison rather than environment effects.

Measures

- **Interface Attention Ratio:** Proportion of visual attention to meeting interface (Document panel)
- **Immediate Recall:** Accuracy on the 12-item memory test
- **Subjective Ratings:** Temporal availability, workload (NASA-TLX), and usability (SUS)

4.3 Results

Cognitive Performance Under Mobility

ScaleStable significantly impaired recall (8.69 items) versus AdaptiveFlow (10.94), AdaptiveLock (10.56), and PeriphGuide (10.25; all $p < .030$). This demonstrates that reduced interface accessibility during mobility directly affected cognitive task performance, creating a clear performance penalty for world-locked approaches under moderate mobility demands.

Visual Attention Patterns During Mobility

ScaleStable yielded significantly less interface viewing time (38.63%) than AdaptiveFlow (54.52%), AdaptiveLock (55.85%), or PeriphGuide (57.79%; all $p < .008$), with a large technique main effect ($F_{3,33} = 12.38$, $p < .001$, $\eta^2 = 0.530$). This confirmed that world-locked anchoring created substantial accessibility limitations during movement phases, forcing users to frequently reposition themselves to access information.

Subjective Experience and Trade-offs

PeriphGuide scored highest on temporal availability (5.71), followed by AdaptiveLock (5.56), AdaptiveFlow (5.50), and ScaleStable (5.08), reflecting users' awareness of accessibility differences. Workload (NASA-TLX) was primarily driven by environment rather than technique: Room to Room (62.88) > Narrow Corridor (48.22) > Open Space (36.74).

However, a striking contradiction emerged in usability preferences. Despite the performance penalty, usability ratings showed AdaptiveLock (83.75) and ScaleStable (80.83) scored highest on SUS, while preference rankings revealed AdaptiveLock/ScaleStable received 92% of first-place votes. This suggests users valued either consistent availability (AdaptiveLock) or the predictability of a fixed spatial reference (ScaleStable), even when accessibility was compromised. Intermediate approaches like AdaptiveFlow and PeriphGuide, despite providing better accessibility, were perceived as less optimal due to persistent user field-of-view interference.

4.4 Study 1 Discussion

Study 1 confirmed that under basic mobility and moderate cognitive demands, **digital information accessibility directly impacts memory performance**. Techniques providing continuous access (body-locked and adaptive) improved objective performance (memory recall) by enabling more consistent information encoding during movement phases.

However, the results revealed an important tension between objective performance and subjective preference. Despite ScaleStable's accessibility limitations leading to poorer memory performance, users still rated it highly for usability. This suggests that users appreciated having a **dedicated, predictable spatial reference** for information, even when it required physical repositioning to access. PeriphGuide, while providing the highest accessibility, suffered from

~~**persistent user field of view interference**—users consistently noted the visual competition between interface content and environmental processing.~~

The absence of technique effects on workload indicates that under moderate multitasking demands, accessibility techniques did not impose significant cognitive costs. However, this raises a critical question: ~~**what happens when environmental complexity increases and users face higher cognitive loads?**~~ Would the accessibility advantages persist, or would the cognitive costs of managing competing visual demands become more pronounced?

~~**5. Study 2: Interface Performance Under Collaborative Cognitive Load**~~

~~**5.1 Motivation and Research Questions**~~

Study 2 examines how anchoring techniques perform when both environmental processing demands and cognitive load increase dramatically. We designed a collaborative sensemaking task requiring constant attention switching between complex physical evidence examination, digital information synthesis, and partner communication.

This context tests whether the accessibility advantages observed in Study 1 persist when users face intensive environmental processing demands (detailed evidence inspection), high cognitive load (collaborative reasoning), and frequent attention switching between multiple information sources. Given Study 1's findings about user preference for spatial predictability, we anticipated that the increased complexity might shift the optimal balance between accessibility and stability.

~~**RQ2:** How does the accessibility-stability trade-off manifest when AR interfaces must support complex environmental processing and collaborative cognitive work?~~

~~**5.2 Method**~~

~~**Participants**~~

12 dyads (24 participants total, 13 female, 11 male, ages 22-38, $M=27.8$, $SD=4.6$) with no prior experience in Study 1.

~~**Task Design**~~

Dyads role-played a murder mystery investigation requiring collaborative sensemaking. The wearer navigated between two physical stations (3.2 m apart) checking evidence consistency, while the remote partner reviewed digital profiles. The task demanded constant attention shifts among:

- ~~**1. Complex environmental processing:** Detailed examination of physical evidence items requiring sustained visual attention~~

2. **Digital information synthesis:** Accessing and processing multiple information sources through the interface
3. **Collaborative reasoning:** Discussing findings, building shared understanding, and coordinating investigation strategy

Dyads created a shared Reasoning Brief including event timeline, suspect identification, evidence items, and conclusion. Five equivalent scripts prevented learning effects.

Procedure

After role assignment and practice, each dyad completed four counterbalanced sessions (one per technique) with questionnaires and interviews. Sessions averaged 23 minutes; full experiments lasted 3–3.5 hours per dyad with remote monitoring.

Measures

- **Dyadic attention coordination:** Interface attention ratio, communicative focus ratio (Video vs. Document attention)
- **Subjective experience:** Perceived attention allocation, workload (NASA-TLX)
- **Collaboration outcomes:** Perceived collaboration quality
- **Qualitative feedback:** Post-experiment interviews on technique trade-offs

5.3 Results

Collaboration Quality and Shared Reference

ScaleStable achieved the highest perceived collaboration quality (6.52) compared to AdaptiveFlow (6.30, $p = .044$) and PeriphGuide (6.32, $p = .004$). This counterintuitive finding—given ScaleStable's accessibility limitations—suggests that **shared spatial stability outweighed individual accessibility benefits** for collaborative performance.

ScaleStable also scored highest on spatial consistency/temporal availability (5.91) compared to AdaptiveFlow (4.32, $p = .011$), indicating that users valued predictable interface positioning for collaborative coordination, even when it required physical repositioning for access.

Attention Coordination Under Complex Environmental Demands

ScaleStable significantly reduced interface attention ratio (25.32%) compared to AdaptiveFlow (40.63%, $p = .003$) or PeriphGuide (40.22%, $p = .003$), with AdaptiveLock intermediate (31.80%). This confirmed that accessibility limitations became even more pronounced when environmental processing required sustained visual attention. However, this reduced interface attention coincided with improved collaboration quality, suggesting that **avoiding user field of view interference** became more valuable than maximizing interface access time.

Cognitive Disruption Costs Under High Load

Unlike Study 1, significant workload differences emerged: AdaptiveLock (38.60) < AdaptiveFlow (58.62, $p = .045$), indicating that **adaptive techniques imposed cognitive costs that became apparent under high task complexity**. Continuous adaptation (AdaptiveFlow) was more disruptive than discrete mode switching (AdaptiveLock), highlighting the importance of transition smoothness.

Perceived attention allocation also favored static techniques: AdaptiveFlow (3.84) < AdaptiveLock (5.13, $p = .012$), with ScaleStable descriptively highest (5.46), suggesting users felt less control over their attention with continuous adaptive behavior and persistent visual interference.

Qualitative Insights: Stability vs. Accessibility Trade-offs

Interviews revealed clear technique preferences based on collaboration needs. ScaleStable's stability aided predictability and shared reference: "The fixed location felt stable, and I always knew where it was (W11)"; "Easier to discuss, stayed aligned on the document (W6)."

AdaptiveFlow created distraction during complex reasoning: "Panel always on the move... chasing it with my eyes (W5)"; "Harder to stay in sync when it kept moving (W12)." PeriphGuide was consistently criticized for visual interference: "Always in my way when trying to examine evidence carefully (W8)." AdaptiveLock provided a middle ground: "Less tiring than the moving one... easier to manage when I needed to focus on evidence (W10)."

5.4 Study 2 Discussion

Study 2 revealed a dramatic context dependent shift in optimal anchoring strategy. Under high collaborative load and complex environmental processing, **ScaleStable's world-locked positioning enhanced collaboration quality despite creating accessibility barriers**. This reversal from Study 1 demonstrates that optimal interface design depends fundamentally on the balance between environmental complexity and cognitive demands.

The findings highlight two critical insights about **visual-cognitive balance under high complexity**:

1. **Environmental processing complexity changes accessibility priorities**: When physical evidence examination required sustained visual attention, the cognitive costs of **user field of view interference** from continuously accessible interfaces outweighed their accessibility benefits. ScaleStable's approach of removing interface content from the primary visual field allowed users to focus on complex environmental processing when needed.
2. **Cognitive disruption costs emerge under high load**: Adaptive techniques that showed no workload penalties in Study 1 created significant cognitive burden when environmental and collaborative demands increased. **AdaptiveFlow's continuous transitions** proved particularly disruptive, creating persistent visual distraction that

~~interfered with both environmental processing and collaborative coordination.~~

5.5 Cross-Study Synthesis: Context-Dependent Optimization

~~Comparing both studies reveals distinct performance patterns for different techniques across varying task contexts:~~

~~**ScaleStable** and **AdaptiveLock** emerged as the most context-appropriate solutions. **AdaptiveLock** achieved higher performance in simpler cognitive multitasking with higher mobility demands (Study 1), while **ScaleStable** performed better in cognitively complex scenarios with moderate mobility demands (Study 2). Both techniques avoid persistent user field of view interference while providing predictable spatial references.~~

~~**PeriphGuide** provides consistent interface accessibility but suffers from **persistent user field of view interference**. While this technique showed reasonable objective performance in Study 1, it was consistently less favored by participants across both studies due to ongoing visual competition between interface content and environmental processing.~~

~~**AdaptiveFlow** demonstrated that **transition smoothness matters critically**. The 2-second interpolation between modes created more disruption than discrete switching (**AdaptiveLock**), particularly under high cognitive load. This suggests that smooth transitions, while seemingly more elegant, can create persistent visual distraction that interferes with both environmental and cognitive processing.~~

Context-Dependent Design Principles:

- ~~● **Moderate Cognitive and Environmental Demands + Higher Mobility** (Study 1): When mobility demands are higher but environmental processing is relatively straightforward, accessibility optimization benefits outweigh user field of view interference and potential cognitive interruption. **AdaptiveLock's** discrete mode switching provides optimal balance.~~
- ~~● **High Cognitive and Environmental Demands + Moderate Mobility** (Study 2): When environmental processing requires sustained attention and cognitive tasks involve collaboration, stability and shared reference advantages outweigh accessibility losses. **ScaleStable's** dedicated spatial positioning supports complex reasoning without visual interference.~~

~~This synthesis suggests that effective mobile AR systems must move beyond simple motion-based adaptation toward **context-aware optimization** that considers environmental complexity, cognitive demands, and mobility requirements. Most critically, designers must carefully balance the competing demands of digital information accessibility against user field of view interference, recognizing that optimal solutions vary dramatically with task context.~~

4. Study 1: Interface Performance Under Basic Mobility Demands

4.1 Motivation and Research Questions

Study 1 investigated interface performance during controlled mobility paired with basic information processing. We examined how anchoring techniques influence individual performance in dual-task scenarios that demand spatial navigation alongside information retention—a typical setup where users balance environmental awareness and interface access.

With simple waypoint navigation and moderate memory recall demands, we predicted that digital information accessibility would be the primary factor affecting memory performance. We also expected trade-offs: accessibility gains from adaptive techniques might introduce cognitive costs during transitions. The task design emphasized continuous movement between spatially distributed waypoints, creating relatively high mobility demands where users frequently needed to access digital information while walking between locations.

RQ1: How do AR information anchoring techniques affect user performance and experience in multitasking with moderate cognitive demand and relatively high mobility demands?

4.2 Method

Participants

We recruited 12 participants (P1–P12; 10 male, 2 female; ages 21–30, $M=24.3$) with normal or corrected-to-normal vision and basic smartphone experience.

Task Design

Participants completed two concurrent tasks to mimic navigating while retrieving collaborative history. In the mobility task (Waypoint Click), four fixed waypoints (labeled 1–4) appeared six times in random order at the bottom of a slide over each 5-minute condition (50 s per interval). Participants walked to the corresponding waypoint via any route and confirmed arrival by clicking a world-locked blue button there. This design created sustained mobility demands as participants continuously moved between distributed locations throughout the session.

The cognitive task (Slides Memory) involved viewing a six-slide deck (50 s per slide) showing two items each—numbers, English words, or simple shapes—in a fixed sequence. After each condition, participants took a 12-item immediate-recall test (4 items per category) to assess how anchoring affected encoding during movement.

Interface Conditions

Participants tested all four techniques (ScaleStable, PeriphGuide, AdaptiveLock, AdaptiveFlow) across three environments (Narrow Corridor, Room-to-Room, Open Space). We

counterbalanced technique order with a Latin square design and fixed environment order to prioritize technique comparisons.

Measures

We tracked interface attention ratio (proportion of visual attention to the document panel), immediate recall accuracy (12-item test), and subjective ratings for temporal availability, workload (NASA-TLX), and usability (SUS).

4.3 Results

Cognitive Performance Under Mobility

ScaleStable significantly impaired recall ($M=8.69$ items) compared to AdaptiveFlow (10.94), AdaptiveLock (10.56), and PeriphGuide (10.25; all $p < .030$). This result aligns with expectations: the other three techniques provide higher accessibility to digital information during movement, while ScaleStable's world-locked positioning restricts access when users are mobile. The performance difference demonstrates that information accessibility directly impacts cognitive task outcomes during continuous mobility.

Validation Through Attention Metrics

Two complementary measures validate this accessibility-performance relationship. First, temporal availability ratings confirmed ScaleStable's limitations: participants rated ScaleStable lowest (5.08) compared to PeriphGuide (5.71), AdaptiveLock (5.56), and AdaptiveFlow (5.50), reflecting their direct experience of reduced access during movement.

Visual Attention Patterns During Mobility

Correspondingly, ScaleStable drew the least interface attention (38.63%) versus AdaptiveFlow (54.52%), AdaptiveLock (55.85%), and PeriphGuide (57.79%; all $p < .008$), with a strong technique effect ($F_{3,33} = 12.38$, $p < .001$, $\eta^2 = 0.530$). This pattern was expected given world-locked designs restrict access during movement, forcing users to reduce information engagement when mobility demands were high.

Subjective Experience and Trade-offs

Workload analysis revealed that anchoring technique itself did not significantly increase cognitive burden—no significant differences emerged between techniques. Instead, workload (NASA-TLX) varied primarily by environment: Room-to-Room (62.88) > Narrow Corridor (48.22) > Open Space (36.74), suggesting that spatial complexity, not interface design, drove cognitive demands.

However, a striking preference pattern emerged that contradicted performance outcomes. Despite ScaleStable's memory performance penalty, it ranked among users' most preferred techniques: AdaptiveLock (SUS=83.75) and ScaleStable (80.83) topped usability ratings, capturing 92% of first-place preference votes. Users favored AdaptiveLock's consistent access or ScaleStable's predictable spatial reference, despite accessibility costs. AdaptiveFlow and PeriphGuide, though more accessible, felt suboptimal due to ongoing field-of-view interference.

Qualitative feedback revealed why users preferred techniques that avoided visual interference. Participants described PeriphGuide as "always in the corner of my eye, distracting from navigation (P3)" and AdaptiveFlow as "constantly moving around, hard to ignore even when I didn't need it (P8)." In contrast, they appreciated ScaleStable's approach: "When I needed it, I knew exactly where to find it. When I didn't, it wasn't bothering me (P11)." AdaptiveLock struck an optimal balance: "Best of both worlds—available when I stopped, out of the way when I moved (P7)."

4.4 Study 1 Discussion

Study 1 demonstrated that during mobility with moderate cognitive demands, **digital information accessibility directly enhances memory performance**. However, users showed a strong preference against persistent field-of-view interference, even when it improved objective performance outcomes.

The results reveal two critical design tensions. **AdaptiveLock** achieved the optimal balance by providing accessibility when needed (body-locked during stationary periods) while minimizing interference during movement (world-locked during mobility). It eliminated the 2-second transition that made AdaptiveFlow feel disruptive, achieving both the best performance and highest user preference.

ScaleStable, despite causing significant accessibility limitations that reduced memory performance, maintained high user preference scores. This suggests that **users highly value minimizing field-of-view interference**, even at the cost of reduced information access. The technique's predictable spatial reference and complete absence of visual distraction during movement resonated with users' desire for control over their visual attention.

These findings raise a critical question: **what happens when task conditions shift?** If environmental complexity increases or mobility demands decrease, would the accessibility-interference trade-off shift toward different optimal solutions? Study 2 investigates how these preferences and performance patterns change under different task constraints.

5. Study 2: Interface Performance Under Collaborative Cognitive Load

5.1 Motivation and Research Questions

Building on Study 1, Study 2 tested anchoring under heightened environmental processing demands and collaborative cognitive load. This sensemaking task required constant attention shifts between detailed physical evidence inspection, digital information synthesis, and partner communication. Importantly, this study also reduced the proportional demand for mobility: while AR wearers moved between investigation stations, they spent significant time in stationary discussion and analysis with their remote partners.

With these shifted demands—complex environmental processing, collaborative reasoning, and reduced mobility emphasis—we investigated how the accessibility-stability trade-off would manifest.

RQ2: How does the accessibility-stability trade-off play out when AR interfaces support complex environmental processing and collaborative cognition?

5.2 Method

Participants

We tested 12 dyads (24 participants; 13 female, 11 male; ages 22–38, $M=27.8$, $SD=4.6$), none from Study 1.

Task Design

Dyads investigated a murder mystery, with AR wearers navigating 3.2 m between physical stations to examine evidence while synthesizing digital profiles and coordinating with remote partners. The task demanded complex environmental inspection (sustained visual focus on physical evidence), digital information synthesis (accessing multiple data sources), and collaborative reasoning (building shared timelines, identifying suspects, reaching conclusions). Five equivalent mystery scripts controlled for learning effects.

Procedure

After role assignment and practice, dyads completed four counterbalanced technique sessions (~23 min each), followed by questionnaires and interviews. Full experimental sessions lasted 3–3.5 hours per dyad under remote monitoring.

Measures

We measured dyadic attention coordination (interface attention ratio; video vs. document focus), subjective attention allocation and workload (NASA-TLX), collaboration quality, and qualitative insights about technique trade-offs from post-session interviews.

5.3 Results

Collaboration Quality and Shared Reference

A striking reversal emerged from Study 1: **ScaleStable achieved the highest collaboration quality** (6.52) compared to AdaptiveFlow (6.30, $p=.044$) and PeriphGuide (6.32, $p=.004$). Its spatial stability apparently provided greater value for collaborative coordination than individual accessibility benefits. ScaleStable also led in spatial consistency ratings (5.91 vs. AdaptiveFlow 4.32, $p=.011$), indicating that predictable interface positioning enhanced team coordination.

Attention Coordination Under Complex Environmental Demands

Paradoxically, this superior collaboration quality was achieved with **less time spent looking at the interface**: ScaleStable users devoted only 25.32% of their attention to the interface versus AdaptiveFlow (40.63%, $p=.003$) and PeriphGuide (40.22%, $p=.003$), with AdaptiveLock intermediate (31.80%). This pattern suggests that when environmental processing demands are

high, **reducing interface attention while maintaining purposeful access** can enhance overall task performance. Users spent more time engaged in focused thinking and evidence analysis rather than managing moving interfaces or dealing with persistent visual interference.

Cognitive Disruption Costs Under High Load

Unlike Study 1, significant workload differences emerged: AdaptiveLock (38.60) imposed less burden than AdaptiveFlow (58.62, $p=.045$). **Continuous adaptation proved more disruptive than discrete mode switching** when cognitive demands increased. Similarly, perceived attention control favored static techniques: participants felt less control with AdaptiveFlow (3.84) compared to AdaptiveLock (5.13, $p=.012$), with ScaleStable achieving the highest sense of attention control (5.46).

Qualitative Insights: Stability vs. Accessibility Trade-offs

Interview feedback highlighted how increased task complexity shifted user priorities toward stability and control. Participants valued ScaleStable's predictability for collaboration: "The fixed location felt stable, and I always knew where it was (W11)"; "Easier to discuss, stayed aligned on the document (W6)."

AdaptiveFlow became particularly problematic under high cognitive load: "Panel always on the move... chasing it with my eyes when I needed to focus on evidence (W5)"; "Harder to coordinate discussion when it kept moving around (W12)." PeriphGuide's persistent interference grew more costly: "Always in my way when trying to examine evidence carefully—couldn't focus properly (W8)." AdaptiveLock provided a middle ground: "Less tiring than the moving one, easier to manage when I needed to focus on evidence (W10)."

5.4 Study 2 Discussion

Study 2 revealed a fundamental shift in optimal interface design when task complexity increased and mobility demands decreased. **ScaleStable emerged as both the best performing and most preferred technique**, representing a complete reversal from Study 1's accessibility-focused results.

Three key factors drove this transformation. **First, complex environmental processing made field-of-view interference more costly.** Unlike Study 1's simple navigation, users needed sustained visual attention to examine physical evidence and make sense of complex spatial relationships. Persistent interface elements (PeriphGuide) or moving displays (AdaptiveFlow) created significant cognitive competition during these critical analysis periods.

Second, collaborative reasoning requires substantial cognitive resources for thinking and coordination. The complex sensemaking task demanded periods of focused reflection and discussion where visual distractions became particularly disruptive. Users needed to allocate cognitive resources to collaborative problem-solving rather than interface management, making ScaleStable's minimal interference approach highly valuable.

Third, reduced mobility emphasis changed the accessibility calculus. While users still moved between investigation stations, the task included substantial stationary periods for discussion and analysis. During these phases, ScaleStable's world-locked positioning was easily accessible, eliminating the accessibility penalty that dominated Study 1's mobile-centric scenarios.

This shift in task demands fundamentally altered the preference ranking between AdaptiveLock and ScaleStable. When mobility was primary (Study 1), AdaptiveLock's adaptive accessibility provided clear advantages. When environmental complexity and collaborative reasoning became primary (Study 2), ScaleStable's stability and minimal interference proved superior. Notably, PeriphGuide and AdaptiveFlow remained least preferred across both studies, highlighting the consistent cost of persistent field-of-view interference regardless of task context.

6. Discussion

6.1 The Dual Attention Challenge in Mobile AR

Our findings reveal the core tension in mobile AR interface <<anchor>> design: users must simultaneously attend to environmental and virtual information while navigating dynamic visual fields. This creates two competing demands: **Environmental Processing** (spatial navigation, object recognition, real-world task execution) and **Virtual Content Access** (information retrieval, interface interaction, digital task completion).

The mobility factor adds critical complexity. As users move, the relative positioning of both environmental and virtual elements changes, requiring frequent visual spatial updating and attention reallocation.

6.2 User Field of View Interference vs. Digital Information Accessibility Trade-offs

****User Field of View Interference Caused by Digital Interface****

The presence of an interface creates environmental viewing interference. Body or head-locked interfaces offer immediate virtual access but create persistent occlusion that becomes problematic when environmental processing demands are frequent and detailed. World-locked positioning with dedicated environmental placement helps manage this central visual competition—users can consciously choose when to shift visual focus based on task needs.

However, world-locked approaches face accessibility challenges during mobility. Depending on viewing distance and angle, content can become unclear or difficult to see. When users move away from the fixed location, the interface may become completely invisible.

****The Adaptive Solution and Its Costs****

Given this trade-off, combining world-locked and body-locked approaches seems logical: world-locked for stationary settings (reducing interference while maintaining accessibility);

~~body locked during movement (solving accessibility problems). However, this introduces~~
~~**cognitive disruption costs**.~~

~~Mode transitions create attention residue effects extending beyond the transition moment itself. Under high cognitive load, these spatial expectation disruptions become significant—each transition requires users to reestablish visual strategies while maintaining complex task state. Conversely, single mode operation better supports cognitive flow states by maintaining consistent spatial-visual expectations, allowing users to develop automatic interface access patterns without disrupting high-level reasoning.~~

~~#### 6.3 Context-Dependent Visual-Cognitive Balance~~

~~Our studies demonstrate that AR interface design may want to consider~~ ****task-specific visual-cognitive balance rather than universal accessibility****~~:~~

~~****Lower Mobility Demands****: With minimal accessibility issues, carefully placed world-locked interfaces provide maximum benefit. Even with some movement, techniques like content size and viewing-angle adjustment introduced in this paper can help maintain accessibility while preserving the advantages of dedicated spatial positioning.~~

~~****Higher Mobility Demands****: Accessibility problems increase, creating greater benefits for strategies that fill accessibility gaps. However, when introducing accessibility techniques, one must consider their potential to interfere with physical content viewing.~~

~~****Adaptive Considerations****: For users alternating between stationary and mobile states, adaptive techniques can be beneficial. However, when users engage in cognitively demanding tasks, adaptive techniques involving mode switching impose cognitive costs that can impede overall performance.~~

~~#### 6.4 Design Implications: Context-Aware Balance Optimization~~

~~Future designs can dynamically balance user field-of-view interference with digital information accessibility against cognitive disruption costs based on:~~

- ~~****Environmental processing complexity**** (navigation demands, visual clutter)~~
- ~~****Cognitive task demands**** (reasoning depth, collaboration requirements)~~
- ~~****User mobility patterns**** (stationary focus vs. active movement)~~
- ~~****Attention switching frequency**** (sustained focus vs. rapid task-switching)~~

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~~This suggests future AR systems can intelligently adapt anchoring strategies based on real-time assessment of environmental complexity, cognitive demands, and user context rather than relying solely on motion-based heuristics.~~

6. Discussion

6.1 The Dual-Attention Challenge in Mobile AR

Our findings reveal an important tension in mobile AR interface design: users often need to simultaneously manage environmental processing (spatial navigation, evidence examination, real-world task execution) and virtual content access (information retrieval, interface interaction, digital task completion) while navigating dynamic visual fields. This dual-attention demand becomes particularly complex during mobility, as the relative positioning of both environmental and virtual elements continuously changes, requiring frequent visual-spatial updating and attention reallocation between competing information sources.

6.2 The Interference-Accessibility Trade-off

Field-of-View Interference vs. Information Access

In mobile AR usage scenarios, digital interfaces inherently create environmental viewing interference. Body-locked and head-locked interfaces provide immediate virtual access but impose persistent visual occlusion that becomes costly when environmental processing demands are frequent and detailed. World-locked positioning with dedicated spatial placement addresses this visual competition by allowing users to consciously control when to shift visual focus based on task priorities.

However, world-locked approaches face accessibility challenges during mobility. Content clarity degrades with viewing distance and angle, and interfaces may become completely invisible when users move beyond optimal viewing ranges. This creates a design tension between minimizing interference and maintaining access.

Adaptive Solutions and Their Hidden Costs

Combining world-locked and body-locked approaches appears to resolve this trade-off through context-sensitive positioning: world-locked during stationary periods (reducing interference while maintaining accessibility) and body-locked during movement (solving mobility access problems). However, our findings reveal that such adaptive approaches can potentially introduce **cognitive disruption costs**. Each spatial transition requires users to reestablish visual strategies while maintaining complex task states, imposing cognitive overhead that can outweigh accessibility benefits. Single-mode techniques (e.g., ScaleStable) can better support sustained cognitive performance by maintaining consistent spatial-visual expectations, allowing users to develop automatic interface access patterns without disrupting higher-level reasoning processes.

6.3 Context-Dependent Optimization

Our studies demonstrate that optimal AR anchoring design requires **task-specific visual-cognitive balance rather than universal accessibility solutions**. The relative importance of interference minimization versus information access shifts based on contextual demands:

High Environmental Processing Demands: When tasks require detailed visual examination or complex spatial reasoning, minimizing field-of-view interference becomes more important. Field-of-view interference can impair environmental processing quality, making world-locked approaches with minimal interference more desirable despite some accessibility limitations.

High Mobility Demands: When continuous movement dominates task requirements, accessibility problems become more significant, creating greater value for techniques that maintain information access during motion. However, any accessibility solution must carefully balance benefits against the costs of persistent field-of-view interference with navigation and environmental awareness.

High Cognitive Load Scenarios: During complex reasoning, collaboration, or sensemaking tasks, cognitive disruption costs become more expensive. Users benefit from predictable, stable interface positioning that supports sustained focus rather than adaptive techniques that require ongoing attention management, even when such adaptation theoretically optimizes an individual factor.

6.4 Design Implications: Context-Aware Interface Strategies

These findings suggest that future AR systems may need to consider **dynamic anchoring strategies** that optimize the interference-accessibility-disruption balance based on real-time task assessment rather than motion-based heuristics alone. Effective context-aware systems can consider:

Environmental Complexity Assessment: Monitor visual clutter, spatial navigation demands, and environmental processing requirements to determine interference tolerance levels. Higher environmental complexity increases the cost of visual occlusion, favoring world-locked positioning.

Cognitive Load Detection: Assess reasoning depth, multitasking demands, and collaboration requirements to evaluate disruption costs. Higher cognitive loads favor stable, single-mode techniques over adaptive approaches that impose transition overhead.

Mobility Pattern Analysis: Track movement frequency, duration, and spatial patterns to determine accessibility needs. Sustained mobility periods increase the value of body-locked access, while mixed mobility-stationary patterns may benefit from carefully implemented adaptive approaches.

Task Phase Recognition: Identify when users are in focused processing modes (detailed examination, complex reasoning, collaborative discussion) versus active transition modes (navigation, information gathering). Stable positioning during focused phases with selective adaptation during transitions can minimize disruption while maintaining necessary access.

User Attention State Monitoring: Detect when users are engaged in sustained environmental processing, virtual content access, or rapid attention switching to optimize interface positioning for current attention allocation patterns.

This context-aware approach suggests that optimal AR interface design lies not in perfecting individual anchoring techniques, but in developing intelligent systems that can dynamically select and configure anchoring strategies based on holistic assessment of user needs, environmental demands, and cognitive state. Such systems could maintain the benefits of specialized techniques while avoiding the costs of inappropriate application across diverse usage contexts.

7. Conclusion

Our work establishes that successful mobile AR interface design requires **contextual optimization of visual cognitive balance** rather than pursuing universal accessibility solutions. The key insight is recognizing when visual field competition costs outweigh accessibility benefits, and when cognitive disruption costs from adaptive strategies exceed their utility.

Future AR systems should move beyond simple motion-based adaptation toward intelligent assessment of the complex interplay between environmental processing demands, cognitive task requirements, and user mobility patterns. The goal is not maximizing digital content accessibility, but optimizing the entire visual cognitive system for effective dual-domain information processing.

7. Conclusion

Our research establishes that effective mobile AR interface design requires **contextual optimization of visual cognitive balance** rather than pursuing universal accessibility solutions. Through two complementary studies examining different task demands and mobility patterns, we demonstrate that optimal anchoring strategies shift dramatically based on the interplay between environmental processing complexity, cognitive load, and movement requirements.

The core contribution lies in revealing when and why different design priorities should dominate: **Field-of-view interference costs can outweigh accessibility benefits** when tasks demand sustained environmental processing, while **cognitive disruption costs from adaptive strategies can exceed their utility** during complex reasoning and collaboration scenarios. These findings challenge the assumption that maximizing digital content accessibility always improves user performance and experience.

Our work provides empirical evidence for a fundamental shift in AR interface design philosophy. Rather than optimizing individual factors like information accessibility or field-of-view interference in isolation, designers should focus on **optimizing the entire visual cognitive system** for effective dual-domain information processing. This requires moving beyond simple motion-based heuristics toward intelligent systems that assess the complex, dynamic relationships between environmental demands, cognitive requirements, and user context.

The practical implications extend beyond academic understanding to real-world AR deployment. As AR systems become more prevalent in professional, educational, and collaborative contexts, the ability to intelligently adapt interface strategies based on holistic task assessment will determine their effectiveness. Our findings provide the foundational understanding needed to build such context-aware systems that truly support users' complex information processing needs rather than simply maximizing technical capabilities.

Future AR interfaces should not ask "how can we make digital content more accessible?" but rather "how can we best support users' complete visual-cognitive workflow?" This research provides the empirical foundation for answering that more nuanced and ultimately more valuable question.

Our research shows that effective mobile AR anchor design demands **contextual optimization of visual-cognitive balance** rather than universal accessibility solutions. Across two studies with varying task demands and mobility patterns, we reveal how optimal anchoring strategies shift based on environmental processing complexity, cognitive load, and movement needs.

Our key contribution identifies when and why specific design priorities should take precedence: **field-of-view interference costs can outweigh accessibility benefits** for tasks requiring sustained environmental processing, while **cognitive disruption from adaptive strategies can exceed their utility** during complex reasoning and collaboration.

We provide empirical support for more considerations in AR anchor design. Designers need to consider optimizing the **entire visual-cognitive system** holistically for dual-domain information processing, moving beyond isolated factors like accessibility or field-of-view interference—and simple motion heuristics—to intelligent systems that account for dynamic environmental demands, cognitive needs, and user context.

Future AR interfaces are encouraged to shift from asking, "How can we make digital content more accessible?" to "How can we best support users' full visual-cognitive workflow?"